LoreWeaver

Dungeons & Dragons

LoreWeaver

Mohan Ye

Yuhui Ni

Jianbin Deng

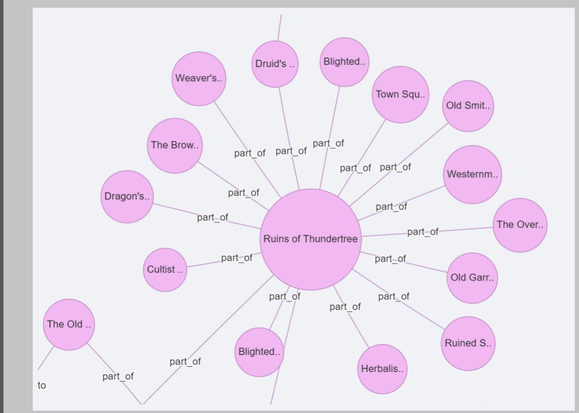
SHANGHAI JIAO TONG UNIVERSITY

Abstract

- We construct LoreWeaver, an automated pipeline that converts D&D 5e modules into heterogeneous knowledge graphs for GraphRAG, enabling AI-powered dungeon masters with context-aware narrative understanding.
- We design a compact knowledge-graph extraction flow that performs multi-phase entity resolution with canonical IDs, infers location hierarchies, links events to participants and places, and captures domain-specific relations (inhabits, guards, hidden_at).
- We implement a dual-pipeline architecture: a large-model path (DeepSeek-Chat, GPT-4) with spatial reasoning, and a small-model path (Qwen3-8B, Llama-3-8B) with XML-tagged extraction, both powering a graph-based RAG engine and an interactive visualization frontend.
- We believe this foundation enables future interactive storytelling, including dynamic quest generation, relationship inference, and immersive world exploration.

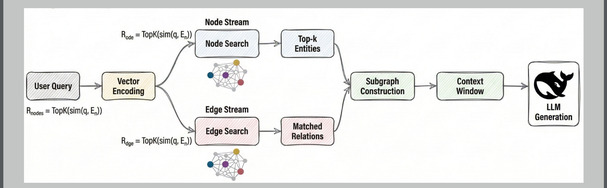
Knowledge Graph Extraction

- Dual-Pipeline Design: The system features a large model pipeline optimized for powerful models (GPT-4, DeepSeek-Chat) and a small model pipeline tailored for resource-constrained parameter models (Qwen3-8B, Llama-3-8B), allowing flexible deployment based on available computational resources.
- Four-Stage Extraction Process: The pipeline sequentially processes adventure data through Shadow Tree construction, Spatial Topology extraction, Section Location mapping, and Entity Relation extraction , with asynchronous concurrent processing at each stage for efficiency.



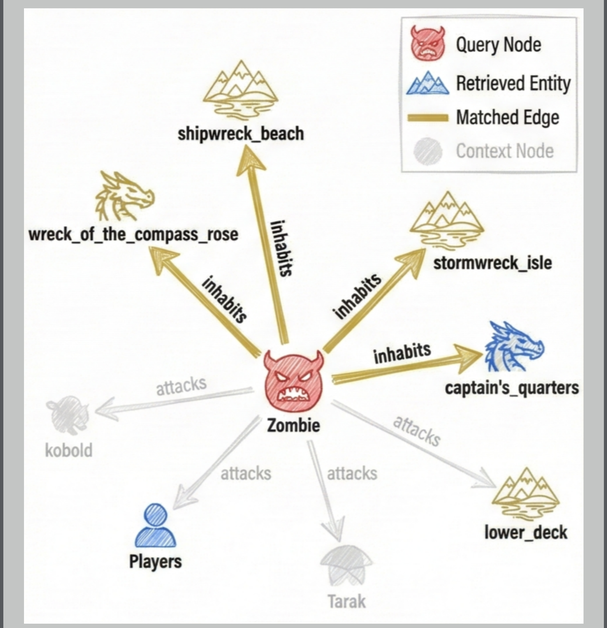
Application: Graph RAG

- Workflow
- Vector Encoding: Pre-compute embeddings for nodes and edges using SentenceTransformer
- Dual Retrieval: Semantic search over nodes and edges
- Context Expansion: Extract k-hop subgraph from matched nodes
- Answer Generation: Feed context to DeepSeek LLM



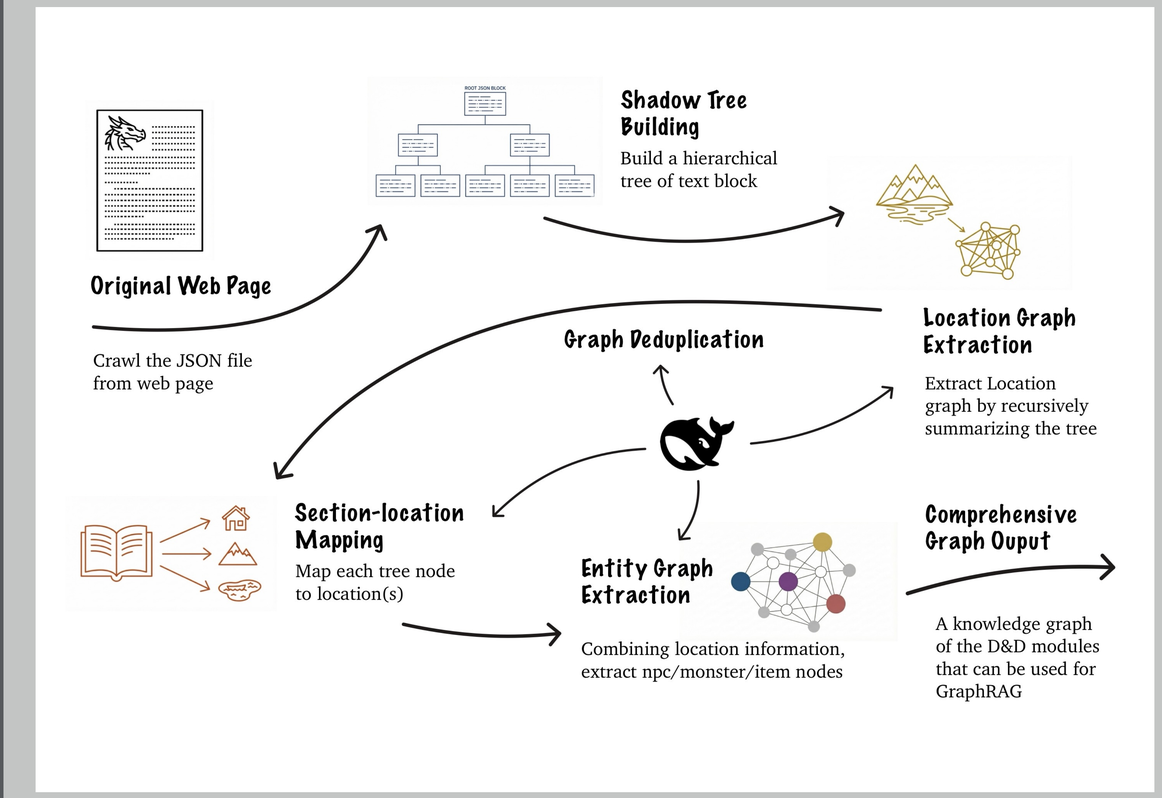
Example: Location Extraction

- Query: "Where are zombies found?"



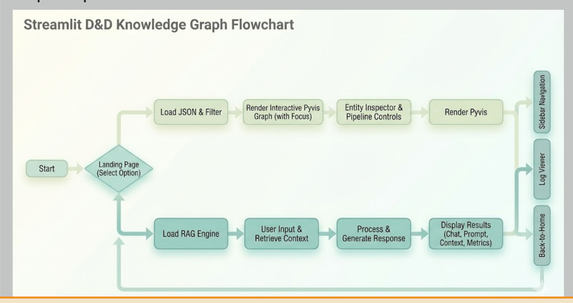
- Entry nodes: zombie, clifftop cemetery, sea caves
- Matched relations:
  - zombies move toward characters and make melee attacks if characters choose to fight
  - some zombies have been trapped in the wreck for years

Process Pipeline

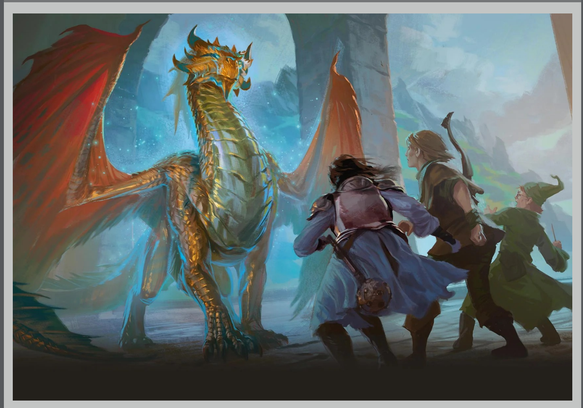


Frontend

- The visualization module loads JSON graph data and renders an interactive network with entity filtering, focus mode, detail inspection, while supporting pipeline .
- The GraphRAG interface retrieves context through graph traversal based on user queries, generates prompts with relevant nodes and edges, and returns answers with full prompt.



Module Example



- The 5etools-formatted JSON serves as the standard input for our pipeline. The figures on the right showcase the Dragons of Stormwreck Isle module, which we use for